

# Adapting CLIK-Diffusion to intraoral scans for tooth alignment prediction

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## 1 Introduction

Orthodontic treatment aims to move the teeth of a patient into a well-aligned position, and planning these geometrical transformations is a time-consuming task that requires expert work to reposition every tooth on a digital 3D model of the dentition. In the last years, several deep learning methods were proposed to automate this step, by predicting the post-orthodontic position of each tooth directly from the pre-orthodontic model. In this work we focus on CLIK-Diffusion by Dou et al. [1], which is a diffusion model for tooth alignment trained on dental models obtained from Cone-Beam Computed Tomography (CBCT), and we adapt it to an IntraOral Scan (IOS) dataset published by Wang et al. in [2].

### 1.1 Related work

Previous learning-based approaches predict, for each tooth, a rigid transformation or a target pose, given the point clouds (or meshes) of the whole dentition. TANet [3] and TAlignNet [4] compute it directly with an MLP, and TAPoseNet [5] does the same with a graph convolutional network. TADPM [6] instead generates the transformation with a diffusion model, whereas TANet+ [7] detects anatomical landmarks and uses them to constrain the orientation of the predicted teeth. CLIK-Diffusion combines these last two ideas and uses a conditional diffusion model to generate the post-orthodontic coordinates of the landmarks. Its pipeline can be divided in three stages:

1. **Landmark detection:** each tooth is represented by a small set of landmarks, which are detected on the tooth mesh by four PointMLP [8] networks, one for each tooth class (i.e., incisors, canines, premolars and molars).
2. **Landmark diffusion:** a conditional diffusion model generates the post-orthodontic coordinates of the landmarks of the whole dentition, as described in **Section 3**.
3. **Rigid transform solving:** for each tooth, the rotation and translation needed to align the initial landmarks to the predicted ones are computed in closed form with the Kabsch algorithm [9], and applied to the tooth mesh.

CLIK-Diffusion is a *clinical-knowledge informed* method, as it uses a standard set of rules for teeth alignment, called *Andrews' six keys to normal occlusion* [10], that are represented through a set of weighted losses computed over mesh transformations. These losses are then added to the mean squared error  $\mathcal{L}_{Base}$  between the ground truth (GT) and the predicted landmarks:

$$\mathcal{L}_{Joint} = \mathcal{L}_{Base} + \lambda_1 \mathcal{L}_{Arch} + \lambda_2 \mathcal{L}_{Inter} + \lambda_3 \mathcal{L}_{Ind} \quad (1)$$

Where:

- $\mathcal{L}_{Arch}$  is the dental arch loss, which fits a fourth-order polynomial to the centers of the teeth of each arch, projected on the horizontal plane, and compares the coefficients of the GT and predicted curves.
- $\mathcal{L}_{Inter}$  is the inter-tooth loss, which penalizes collisions between neighboring teeth.
- $\mathcal{L}_{Ind}$  is the individual tooth loss, computed as the cosine similarity between the predicted and ground truth orientations of each tooth.

## 1.2 Task formulation

CLIK-Diffusion’s repository<sup>1</sup> provides the pre-trained weights for its landmarker and its diffusion model, trained on meshes extracted from CBCT data. Thus, the goals of this project are:

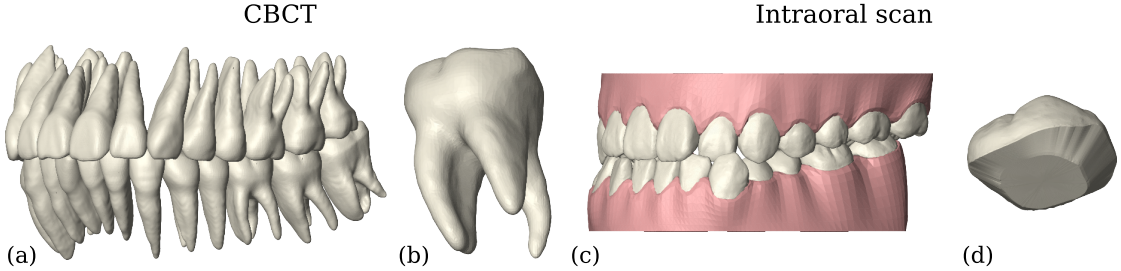
1. Adapt the inference pipeline of CLIK-Diffusion to the IOS dataset, by preprocessing the scans so that they match the input format and coordinate frame expected by the model, and evaluate its performance on new data.
2. Finetune the diffusion model on IOS data, and compare this adaptation with the one proposed in the original paper, where the landmark detection networks are retrained instead.
3. Analyze the model performance with respect to specific clinical labels (i.e., malocclusion, overbite and overjet), to understand whether some kinds of patients are harder to predict.

Each patient is represented by the crown meshes of its initial dentition  $X = \{X_i\}_{i=1}^{28}$ , where  $X_i$  is the  $i$ -th tooth, and the pipeline produces in output a rigid transformation for each tooth, so that the predicted arrangement is  $\hat{X}_i = R_i X_i + T_i$ . More formally, the model  $h$  we want to learn can be expressed as follows:

$$h : X \rightarrow Y, X = \{X_i\}_{i=1}^{28}, Y = \{(R_i, T_i)\}_{i=1}^{28}, R_i \in SO(3), T_i \in \mathbb{R}^3$$

## 2 Data preprocessing

The dataset used in this work, published by Wang et al. in [2], consists of 1060 pairs of pre- and post-orthodontic 3D dental models acquired with an intraoral scanner, where the post-orthodontic models are generated by dental technicians. Each model consists of upper and lower arch meshes, together with segmentation data that maps the vertices of each tooth to its FDI<sup>2</sup> number. Additionally, each patient is labelled with a set of clinical categories (i.e., malocclusion, overbite and overjet), which are used in **Section 4.1** to stratify the results.



**Figure 1:** CBCT scan of full dentition (a), and single molar (b), versus an IOS scan (c) where the base of each tooth is automatically joined to create a closed mesh (d).

CBCT data is a volumetric X-ray imaging technique that can also capture the roots hidden inside the bone, while an intraoral scanner is only capable of capturing the surfaces visible inside the mouth, thus IOS meshes contain crowns and gingiva but no roots, and the surfaces it cannot reach (i.e., the contacts between neighboring teeth and below the gingiva) are filled in automatically to create a closed mesh, as shown in **Figure 1**.

Since each scan has to be preprocessed to match the input format expected by CLIK-Diffusion, the following steps are applied:

**Axis alignment:** the rotation  $R$  swaps the left-right and anterior-posterior axes of the IOS scans and flips their vertical axis, to match CLIK’s coordinate frame:

$$R = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

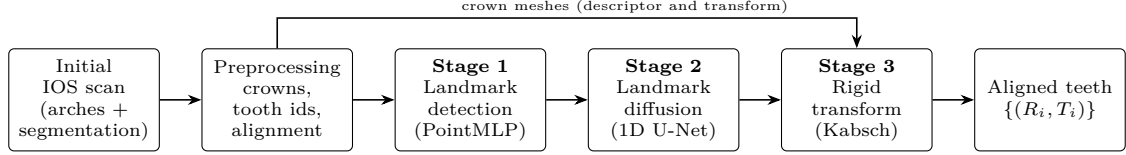
<sup>1</sup><https://github.com/ShanghaiTech-IMPACT/CLIK-Diffusion>

<sup>2</sup>[https://en.wikipedia.org/wiki/FDI\\_World\\_Dental\\_Federation\\_notation](https://en.wikipedia.org/wiki/FDI_World_Dental_Federation_notation)

**Translation:** each arch mesh is recentered by subtracting the mean of both arches’ vertices, so that the origin of the coordinate system is at the center of the dentition.

Finally, 158 patients whose initial and final scans do not contain the same set of teeth (due to extractions) are discarded, which leaves 902 usable patients, thus all the results of this work refer to non-extraction cases only.

### 3 Model architecture



**Figure 2:** High level overview of the adapted pipeline

As shown in **Figure 2**, the crowns obtained from the preprocessing step are passed through the three stages of CLIK-Diffusion, which are reused without changes from their original implementation, apart from the weights of the diffusion model.

In stage 1, the landmarks of each crown are detected by the PointMLP network [8] of its tooth class and, in total, we obtain a set of 256 landmarks for the whole dentition.

In stage 2, the landmarks are diffused using a 1D U-Net [11], which learns to generate new landmark positions by reversing a noising process.

Stage 3 computes the rigid transformation ( $R, T$ ) needed to align the source tooth with the target configuration.

Diffusion models, learn to generate data by reversing a gradual noising process: during training, Gaussian noise is added to the target landmarks and the network learns to remove it, so that, at inference time, the post-orthodontic landmarks can be generated by iteratively denoising pure noise. In CLIK-Diffusion the denoising network is conditioned on two inputs:

- **Landmark condition**  $L \in \mathbb{R}^{256 \times 5}$ : the normalized coordinates of each landmark ( $L_{init} \in \mathbb{R}^{256 \times 3}$ ), concatenated with landmark and tooth IDs ( $L_{id} \in \mathbb{R}^{256 \times 2}$ ).
- **Dentition descriptor**  $D \in \mathbb{R}^{256 \times 128 \times 3}$ : around each landmark, a set of 128 points are sampled in a 2mm radius, which describe the local geometry of the tooth for each landmark.

To evaluate a prediction, the GT mesh is preprocessed with the same rotation and translation as the initial one, and we use Kabsch’s algorithm, as in Stage 3, to compute the rigid transformations. Successively, the predictions are compared with the GT using the four metrics of [1], which are computed for each tooth and then averaged over the whole dentition:

- **Rotation error:** the angle between the GT and predicted rotations of the tooth, in degrees.
- **Translation error:** the mean absolute difference between the GT and predicted translations.
- **Point cloud error:** the mean distance between corresponding vertices of the predicted and GT crowns.
- **Collision error:** the mean penetration depth of each tooth into its neighbors.

#### 3.1 Diffusion model finetuning

To finetune the diffusion model, we keep the landmarker weights frozen and use a standard supervised learning pipeline on our dataset. The 902 scans are split in training, validation and test sets of respectively 722, 90 and 90 patients, while ensuring a correct stratification of the clinical classes is maintained.

A minimal grid search was performed to find a good configuration of hyperparameters and loss function. For example, instead of using a simple MSE loss, we achieved better results by using a loss

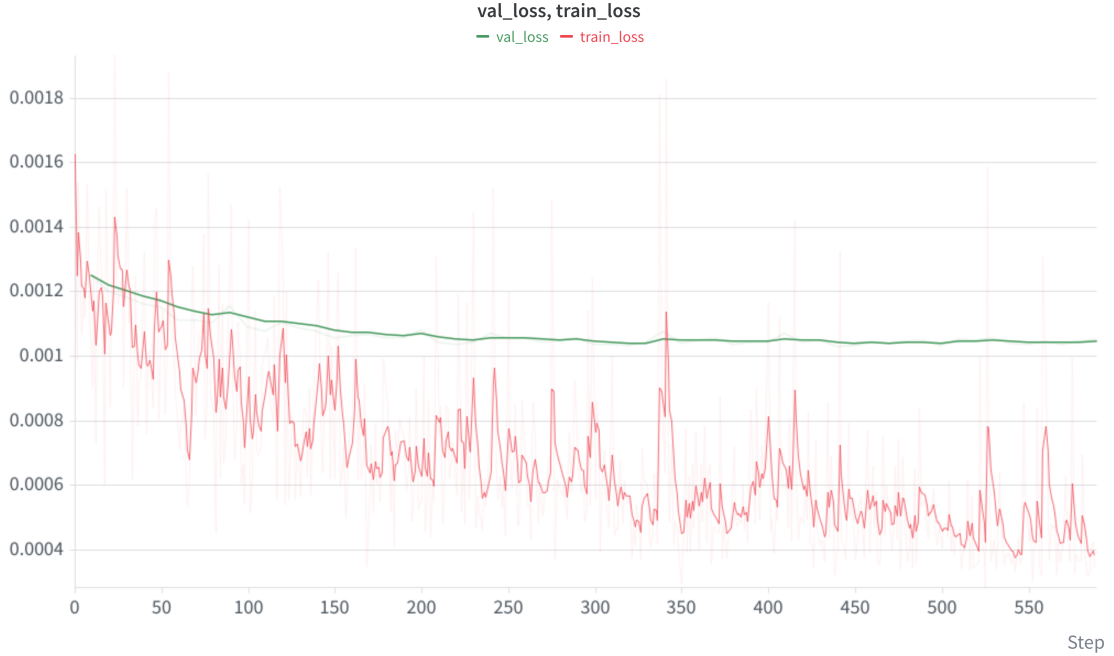
function similar to Eq.1, which is a weighted sum of the base loss and the two clinical-knowledge informed losses:

$$\mathcal{L} = \mathcal{L}_{Base} + \lambda_1 \mathcal{L}_{Arch} + \lambda_2 \mathcal{L}_{Ind} \quad (2)$$

The inter-tooth loss  $\mathcal{L}_{Inter}$  was not used for computational reasons, as it would have required to compute the collision loss for every pair of neighboring teeth at every training step, leading to a significant increase in training time.

The hyperparameters (HPs) used for finetuning are the following:

- **Optimizer:** AdamW, without weight decay.
- **Learning rate:**  $\eta = 1 \times 10^{-5}$ , constant.
- **Batch size:** 16 patients.
- **Loss weights:**  $\lambda_1 = 5 \times 10^{-5}$  and  $\lambda_2 = 0.01$ , chosen so that the three terms of Eq.(2) have the same order of magnitude.
- **Early stopping:** with a patience of 15 epochs on the validation loss, after which the weights with the lowest validation loss are restored.



**Figure 3:** Training and validation loss curves during finetuning.

On average, the finetuning process required about 45 epochs to converge, with a training time of 9 minutes on a single RTX 2080 GPU.

## 4 Experimental results

This section compares three ways of running CLIK-Diffusion on intraoral scans:

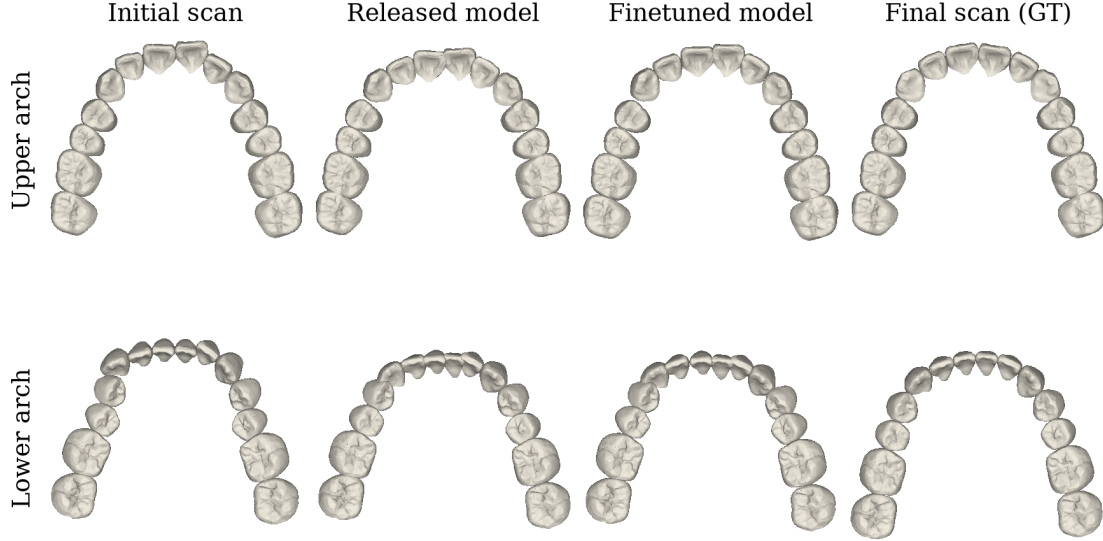
- **Released weights:** trained on CBCT data only, without any adaptation.
- **Retrained landmarker:** the data reported from Table 2 of CLIK’s paper [1], where the authors retrained the landmark detection networks on 100 IOS models annotated by their dentists.
- **Finetuned diffusion (ours)**

| Method                     | Rotation<br>(°)                     | Translation<br>(mm)                 | Point<br>Cloud (mm)                 | Collision<br>(mm)                   |
|----------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| TANet [3]                  | $11.685 \pm 3.189$                  | $2.505 \pm 0.624$                   | $3.593 \pm 1.307$                   | $0.609 \pm 0.205$                   |
| TANet+ [7]                 | $10.878 \pm 3.028$                  | $2.173 \pm 0.551$                   | $2.878 \pm 1.086$                   | $0.696 \pm 0.157$                   |
| TADPM [6]                  | $9.807 \pm 3.782$                   | $1.765 \pm 0.678$                   | $1.543 \pm 0.722$                   | $0.194 \pm 0.129$                   |
| <i>CLIK-Diffusion</i>      |                                     |                                     |                                     |                                     |
| Released weights           | $8.732 \pm 2.520$                   | $1.750 \pm 0.467$                   | $1.591 \pm 0.708$                   | $0.257 \pm 0.112$                   |
| Retrained landmarker       | <b><math>6.688 \pm 2.561</math></b> | <b><math>1.199 \pm 0.441</math></b> | <b><math>1.295 \pm 0.678</math></b> | <b><math>0.168 \pm 0.128</math></b> |
| Finetuned diffusion (ours) | $7.582 \pm 2.709$                   | $1.495 \pm 0.536$                   | $1.300 \pm 0.654$                   | $0.193 \pm 0.109$                   |

**Table 1:** Other methods compared to our finetuned model and the author’s one.

**Table 1** extends Table 2 of CLIK-Diffusion’s paper, which compares the finetuned model with previous methods on the IOS dataset.

Both adaptations outperform the released weights and the previous methods, and reach almost the same point cloud error, whereas retraining the landmarker is generally more effective on the other metrics.



**Figure 4:** Qualitative comparison of a test patient.

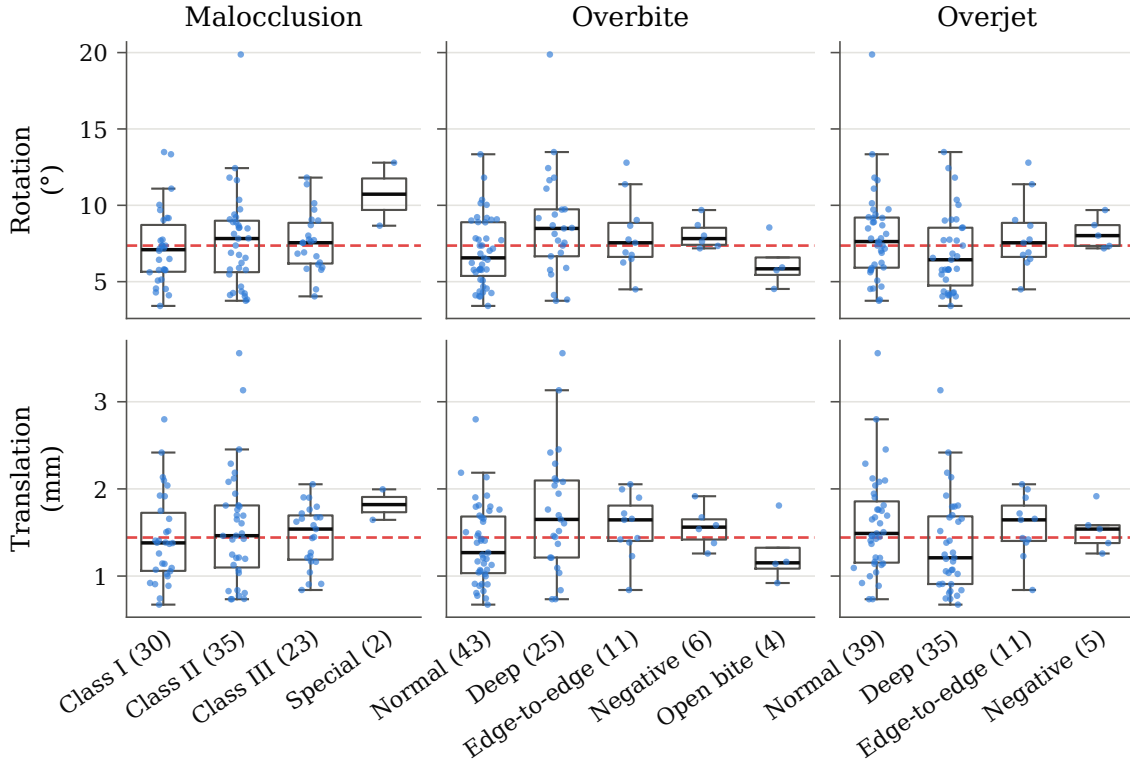
#### 4.1 Results by clinical class

The dataset labels each patient with its malocclusion class (i.e., Class I, II, III or special), and with its overbite and overjet categories, which describe respectively the vertical and horizontal overlap of the incisors (e.g., an overjet between 0 and 3mm is normal, above 3mm is deep and below 0mm is negative).

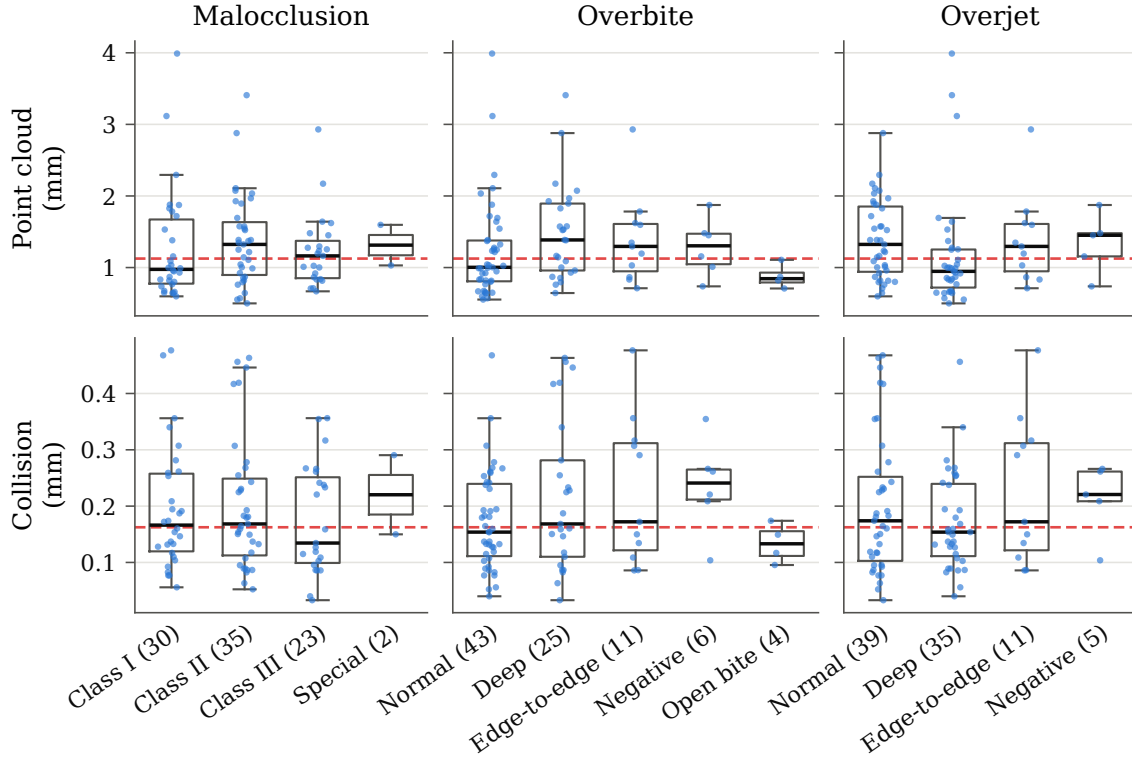
**Table 2** shows how finetuning reduces the mean PC error in every class by an average of 21.7%, and overall the model does not appear to perform significantly better or worse on any specific class, as shown qualitatively in **Figure 5** and **Figure 6**.

| Class               | n  | Released<br>PC (mm) | Finetuned<br>PC (mm) | $\Delta$ PC<br>(mm) |
|---------------------|----|---------------------|----------------------|---------------------|
| <b>Malocclusion</b> |    |                     |                      |                     |
| Class I             | 30 | $1.61 \pm 0.85$     | $1.27 \pm 0.78$      | $-0.34$ (−20.8%)    |
| Class II            | 35 | $1.56 \pm 0.61$     | $1.37 \pm 0.64$      | $-0.19$ (−12.1%)    |
| Class III           | 23 | $1.58 \pm 0.69$     | $1.22 \pm 0.52$      | $-0.36$ (−22.5%)    |
| Special             | 2  | $2.00 \pm 0.01$     | $1.31 \pm 0.40$      | $-0.69$ (−34.3%)    |
| <b>Overbite</b>     |    |                     |                      |                     |
| Normal overbite     | 43 | $1.52 \pm 0.73$     | $1.23 \pm 0.69$      | $-0.29$ (−19.1%)    |
| Deep overbite       | 25 | $1.67 \pm 0.65$     | $1.49 \pm 0.67$      | $-0.17$ (−10.4%)    |
| Edge-to-edge bite   | 11 | $1.66 \pm 0.58$     | $1.38 \pm 0.62$      | $-0.28$ (−17.0%)    |
| Negative overbite   | 6  | $2.07 \pm 1.02$     | $1.28 \pm 0.40$      | $-0.79$ (−38.0%)    |
| Anterior open bite  | 4  | $1.23 \pm 0.23$     | $0.88 \pm 0.17$      | $-0.35$ (−28.7%)    |
| <b>Overjet</b>      |    |                     |                      |                     |
| Normal overjet      | 39 | $1.62 \pm 0.66$     | $1.39 \pm 0.55$      | $-0.23$ (−14.5%)    |
| Deep overjet        | 35 | $1.46 \pm 0.71$     | $1.17 \pm 0.79$      | $-0.29$ (−19.7%)    |
| Edge-to-edge        | 11 | $1.66 \pm 0.58$     | $1.38 \pm 0.62$      | $-0.28$ (−17.0%)    |
| Negative overjet    | 5  | $2.12 \pm 1.13$     | $1.34 \pm 0.42$      | $-0.78$ (−36.8%)    |

**Table 2:** Point cloud error of the released and finetuned models for each clinical class, and change of the mean error brought by finetuning ( $\Delta$ PC).



**Figure 5:** Rotation and translation errors of the finetuned model grouped by clinical class. Each blue dot is a patient, and the red line is the median error.



**Figure 6:** Point cloud and collision errors, as in **Figure 5**.

## 5 Conclusions

The adaptation of CLIK-Diffusion to intraoral scans shows that finetuning the diffusion model is a viable alternative to retraining the landmarker, which would require having a panel of dentists annotate a large number of scans. The finetuned model outperforms the released weights as well as previous methods, and reaches almost the same point cloud error as the retrained landmarker, while being marginally worse on the other metrics. The analysis by clinical class shows that the model does not perform significantly better or worse on any specific class.

## References

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